OMKAR KADAM

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EDUCATION

Indian Institute of Technology, Patna *2025 - 2027*

M.Tech in Artificial Intelligence | CPI: 7.51/10 | Patna, Bihar

Marathwada Mitra Mandal College of Engineering *2021 - 2025*

B.E. in Computer Engineering | CPI: 7.69/10 | Pune, Maharashtra

PROJECTS

Self-Critiquing RAG System with LangGraph | GitHub *May 2026*

- Engineered an end-to-end agentic RAG pipeline ingesting **3 research PDFs (72 pages)** into **317 structure-aware chunks** for semantic retrieval using Gemini embeddings and FAISS.
- Designed a cyclical LangGraph workflow integrating **4 LLM agent components**—retrieval critic, query reformulator, answer generator, and answer critic—with iterative retrieval and configurable retry limits.
- Evaluated retrieval across **20 benchmark questions**, achieving **95.83% Recall@10** and **81.42% MRR** with structure-aware chunking and FAISS retrieval.
- Implemented evidence sufficiency checks, source/page-level citations, and answer-groundedness auditing; verified runtime execution of the self-critique loop with **3 retrieval attempts** and **2 reformulations** on a challenging RAG query.

Tools & Techniques: Python, LangGraph, LangChain, Google Gemini, FastAPI, Pydantic

LLM Guardrails Gateway | GitHub *January 2026*

- Developed an **LLM Guardrails Framework** using Python and FastAPI to detect prompt injection, PII leakage, toxicity, and policy violations in LLM inputs and outputs.
- Built a benchmark comprising **500 labeled test cases across 9 guardrail categories**, achieving **94.2% classification accuracy** and **88.47% macro F1-score**.
- Implemented an evaluation engine with confusion matrices, precision, recall, F1-score, false-positive/negative rates, and latency metrics for systematic guardrail assessment.
- Integrated benchmark results into an interactive web dashboard and validated the application with **47 passing unit and integration tests**.

Tools & Techniques: Python, FastAPI, Google Gemini, PyYAML, Pydantic, jsonschema

End-to-End Phishing Detection System (MLOps) | GitHub *August 2025*

- Engineered an end-to-end **MLOps pipeline for phishing detection**, integrating data ingestion, validation, drift detection, preprocessing, model training, and prediction using MongoDB and FastAPI.
- Evaluated multiple ML classifiers using scikit-learn and automated experiment tracking with **MLflow and DagsHub**, selecting **Random Forest (256 estimators)** with **97.38% test accuracy, 97.02% phishing-class F1-score, and 99.78% ROC-AUC** in the reported evaluation.
- Containerized the application with **Docker** and implemented **GitHub Actions CI/CD** to build and push images to AWS ECR, deploy to AWS EC2 via a self-hosted runner, and store artifacts in AWS S3.

Tools & Techniques: Python, scikit-learn, FastAPI, MLflow, DagsHub, MongoDB, Docker, GitHub Actions, AWS EC2, AWS ECR, AWS S3

SKILLS

Programming & Web: Python, C/C++, SQL, JavaScript, HTML, CSS

Machine Learning & NLP: Scikit-learn, PyTorch, TensorFlow, CNNs, Transformers, TF-IDF, Feature Engineering, Cross-Validation

LLM & Agentic AI: LangGraph, LangChain, Google Gemini, RAG Pipelines, Prompt Engineering, Guardrails

MLOps & Tools: MLflow, DagsHub, FastAPI, Docker, GitHub Actions, AWS EC2, AWS ECR, AWS S3, Git, Linux, MongoDB, Pydantic, PyYAML, Containerization, Drift Detection

ACHIEVEMENTS

- Secured **All India Rank 1603** in GATE 2025 (Top ~1% among ~1.7 lakh candidates)
- Solved **400+ problems** on LeetCode. (Profile)
- Achieved a **3-star rating** on CodeChef; ranked **438 of 8,000** in CodeChef Contest 245(Profile)

CERTIFICATIONS

- Udemy - C/C++ Certification
- Udemy - Data Science Certification
- Udemy - Data Structures and Algorithms Certification